HAVN Rest Cabins

Health, sleep and longevity concept. Technical summary for Soundbox Inc., Guangzhou. HAVN, Berlin, 21 October 2026. A basis for discussion, not a specification release: every point can be adapted together.

1. Cabin types

Every HAVN station combines three cabin types around a shared sanitary core (showers, WC, lockers). All three use the same panel system and the same soundproof door: 2.40 m interior height, flat-pack panels max. 80 cm wide, the Soundbox air-conditioning unit on the roof with a total outer height of max. 2.70 m including the unit, 230 V EU power, CE, reaction-to-fire class per EN 13501-1.

Type	Floor (m)	Interior height	Bed / interior	Use
Single Cabin	2.0 x 2.4	2.40 m	Bed 200 x 100 cm, small desk and chair, hooks, luggage space under the bed	Naps, quiet work breaks, overnight stays
Double Cabin	2.4 x 2.4	2.40 m	Bed 200 x 160 cm, hooks and luggage space for two	Same as Single, for two
Health Pod	2.0 x 2.4	2.40 m	Normal bed 200 x 100 cm exactly like the Single Cabin, plus wall-mounted red/NIR light panel (630-850 nm), audio and scent	Ordinary overnight stays like every cabin, plus guided 20-minute protocols

2. Design principle: four measurable conditions plus one light sequence

The health value of a rest cabin lies in conditions that can be measured and documented per stay: darkness, quiet, fresh air and the right temperature, plus a disciplined light sequence (warm low light in the evening, true dark at night, a 30-minute sunrise before waking). These levers carry the strongest evidence, they are inexpensive, and the best sleep operators (Zedwell London, Equinox New York) sell exactly this set. Everything else is comfort or an opt-in extra.

Parameter	HAVN target	Basis
Noise at the pillow	LAeq,night at most 30 dB(A). Envelope: wall R'w at least 52 dB, door Rw at least 50 dB; silencers on the air-conditioning penetrations; no audible fan at low speed; measured dB(A) at the pillow to be published	WHO bedroom guideline; DIN 4109. Office pods rated under ISO 23351-1 reach about 30 dB; a station site needs roughly 10 dB more
Air and climate	Soundbox air-conditioning unit with heating, cooling and air purification, quiet version with sound-wave cancelling. Guest setpoint from 16 °C (expected 16-26 °C), default 20 °C. Fresh-air supply controlled by CO2: boost at about 750 ppm, below 1,000 ppm all night. HEPA + activated carbon; 40-60 % RH; full air change during the 20-minute reset. No ioniser, plasma or ozone device. Defined heat and condensate path into the host building	EN 16798-1 (about 36 m3/h fresh air per person); DTU chamber studies (sleep efficiency and awakenings worsen above 1,000 ppm); Baniassadi et al. 2023; WELL v2
Light during sleep	Below 1 lux at the pillow; all status LEDs off; hard-off relay for the virtual window and the control tablet; floor cue lines only on bed exit	Brown et al. 2022 consensus; PNAS 2022 (100 lux at night raises heart rate and next-day insulin resistance)
Particulates	PM2.5 at most 10 ug/m3 on the supply air	WELL v2 A05

3. Lighting: RGB+CCT with true white, and the sunrise specification

RGBW with true white: red, green, blue plus warm-white and cool-white channels (RGB+CCT), so that white always comes from the white channels and never from an RGB mix; tunable 2200-6500 K, CRI 90 or higher; flicker-free drivers (PstLM at most 1.0, SVM at most 0.4 as in EU Ecodesign 2019/2020); scenes executed locally on the cabin controller. Control: any controller or software with an open, documented API that the HAVN app can use (DALI, Casambi, Zigbee, Matter or a proprietary controller are all acceptable). HAVN asks for the measured spectral power distribution and the computed melanopic EDI per channel. Every light source needs an EN 62471 photobiological report. Floor orientation lines are not lit permanently: they switch on in dim red/amber (5-20 lux at floor level) only when the bed-exit sensor detects feet on the floor, and switch off when the guest is back in bed or after two minutes.

Scene	Melanopic EDI at the eye	CCT	Photopic lux at the eye
Daytime arrival, reading, nap exit	at least 250 lx	4000-6500 K	about 265-350 lx
Evening arrival (last 3 h before sleep)	at most 10 lx	2200-2700 K	about 20 lx ambient, warm localised reading light
Wind-down (last 20-30 min)	1-5 lx	amber, below 2200 K	a few lux

Scene	Melanopic EDI at the eye	CCT	Photopic lux at the eye
Sleep	at most 1 lx	-	below 1 lx; window and tablet hard-off
Night navigation (bed exit only)	at most 10 lx	deep red / amber at floor level	5-20 lx of long-wavelength light as floor cue lines
Sunrise (see below)	ramp to 180-250 lx	1800-2000 K to 4500-5000 K	0 to 250-300 lx over 30 minutes, starting before the alarm

Sunrise specification. Starts 30 minutes before the wake-up time from complete darkness; perceptually smooth dimming curve (logarithmic, no visible steps); white only from the true-white channels. For the 20-minute nap the same sequence is compressed to about five minutes. The evening wind-down runs in reverse over 20-30 minutes (2700 K to 2200 K to amber, about 20 lux to below 1 lux at the eye).

Minute	Colour temperature	Light at the eye	Channels
0-8	amber/red, equivalent to about 1800-2000 K	0 to 5 lux	red and amber only, white channels off
8-16	2000 to 2700 K	5 to 40 lux	warm white fades in
16-24	2700 to 3500 K	40 to 120 lux	warm white, cool white starts
24-30	3500 to 4500-5000 K	120 to 250-300 lux (about 180-250 melanopic lux)	cool white dominant; wake-up sound rises in the last two to three minutes
after wake-up	5000-6500 K	at least 250 melanopic lux (about 300-350 lux) for 5-15 minutes	"daylight" for getting up

4. The virtual window

HAVN cabins have no real window. Randomised studies show that an artificial window or skylight lowers tension, anxiety and claustrophobia and improves mood in windowless rooms; no study yet covers sleeping rooms, so HAVN will measure this in its own cabins. Depth is the decisive filter for a 100 mm panel:

Technology / product	Depth	Power	Fits a 100 mm wall?
Innerscene Circadian Sky (tunable sky, no sun), 300 x 600 to 600 x 1200 mm	84 mm	35-75 W	Yes, flush
Edge-lit "virtual window" / skylight light box (Sky Factory type, Chinese OEM), 450 x 1200 or 600 x 1200 mm	about 45-85 mm	tens of W	Yes, surface or flush; also ceiling
Thin LCD window (e.g. Atmoph 27")	30-60 mm	n/a	Yes
Rayleigh sun-and-sky units (Yeelight Pro P21, Mitsubishi Misola, Innerscene A7, CoeLux HT25)	120-270 mm	80-200 W	No: roof cut-out with a housing above the cabin, only where the host ceiling leaves that room above the 2.70 m

- A portrait 450 x 1200 mm panel fits inside one 80 cm wall module; a landscape 1200 x 600 spans two modules and puts a panel joint behind the luminaire.
- Recessing an 84 mm luminaire into a 100 mm acoustic sandwich removes mass and absorber over that area and creates an acoustic weak spot. HAVN therefore prefers the ceiling panel (where the "outside" is the host room) or a surface-mounted 46 mm light box on the wall.
- The window runs on the same lighting control: daytime light dose, sunrise in the morning, hard-off during the sleep window. Its heat must be carried by the air stream of the air-conditioning unit.
- For EU sale an OEM panel needs CE (Low Voltage and EMC), RoHS, an EN 62471 report and flicker data (PstLM, SVM).

5. The Health Pod

The same cabin with the same normal bed, so it is used for ordinary overnight stays like every other cabin. In addition: a wall-mounted red/near-infrared panel, audio and scent, and guided protocols. Positioning: wellness and recovery support, explicitly not a medical device. HAVN will not make skin, sleep-disorder, disease or "longevity" claims for the panel; the best-supported effects of red/NIR light are muscle recovery, mood and alertness, so sessions are scheduled in the daytime, not before sleep.

- Protocols: 20-minute power nap (settling with a breathing cue, near-dark, lying time capped, wake by rising audio and the compressed sunrise, 2-3 minutes of bright "re-boot" light before exit); 10-minute resonance-breathing or guided rest audio; red-light session of about 10 minutes followed by 10 minutes of rest; 90-minute full sleep cycle with a soft movement-based wake window.

– Panel engineering: half-body panel (about 91 x 30 cm, typically 300-500 W) fixed on the long wall beside the bed at 20-30 cm; nine-point irradiance map at mounting distance from a calibrated spectroradiometer; EN 62471 report including the separate retinal-thermal limit for 780-1400 nm; session timer and per-booking exposure lock; thermal cut-out; disposable eye protection; no flashing or strobe modes. From the panel system: mounting frame at the defined distance and cabling.
– Heat: a 450 W panel running 15 minutes releases about 400 kJ into the cabin. The air-conditioning unit must run at full capacity during and after a session, and the fan noise of the panel must be specified.
– Optional later modules: vibroacoustic transducers in the bed, far-infrared warm mat with surface temperature capped around 40-45 °C, massage bed. Pulsed electromagnetic field mats, compression boots and cold exposure are not planned for unattended use.

6. Sound, voice, sensing and controls

– Two speakers per cabin (stereo) with a clean, high-quality sound: nothing extravagant, simply a clear and pleasant reproduction of voice, soundscapes, pink/brown masking noise (default off) and guided breathing or "non-sleep deep rest" audio.
– One microphone per cabin for snoring detection in the sleep report and for voice control. Speakers and microphone are wired to the cabin controller with an audio interface or API. Privacy: hardware mute with an indicator for the microphone, audio processed locally wherever possible, no cameras.
– HAVN will add an AI voice assistant on this hardware: guests set light, temperature, sound and wake-up time by voice and ask questions about their stay and sleep. The cabin only needs the audio hardware and the open API; the assistant comes from HAVN.
– Sensors: one contactless 60 GHz radar module per cabin (breathing, heart rate, movement, bed occupancy and bed exit; triggers the floor lights and confirms the cabin is empty for the reset); CO2/temperature/humidity (e.g. Sensirion SCD41), particulates (SPS30), sound level, light level. Per-stay environment record (CO2, temperature, dB) stored by the cabin controller.
– Controller: local scene execution (light, air-conditioning, audio, scent, window), no cloud dependency, open and documented API to the HAVN app and booking system; platform to be agreed (DALI, Casambi, Tuya Zigbee, ESP32-based, Matter or proprietary).
– Guest-facing functions: optional sleep report after the stay (opt-in per stay), soft wake window in light sleep, in-app health screening before Health Pod modules. Data processing follows EU GDPR: tracking off by default, explicit consent, pseudonymised stay IDs, EU hosting.

7. Materials, hygiene and compliance checklist

These are the documents HAVN needs from the cabin manufacturer for the EU market. Sustainability matters to HAVN: the cabin should be healthy for the guest and for the environment, so its ecological properties and pollutant emissions will be tested independently.

Area	Requirement	Standard / rule
Formaldehyde	Board materials ENF or E0 (GB/T 39600-2021). Chamber test report on the finished cabin article at or below 0.062 mg/m3 from an ISO 17025 laboratory. Chinese E1 boards (0.124 mg/m3) no longer meet the EU limit in force since 6 August 2026	REACH Annex XVII entry 77, Appendix 14
VOC emissions	28-day emission test on the assembled wall, ceiling and floor build-ups (not on raw boards); REACH SVHC declaration for foams, adhesives and textiles	German AgBB scheme, 2024 edition
Reaction to fire	Classification reports from an EU-accredited laboratory (GB 8624 reports are not accepted in Germany): B-s1,d0 for every exposed lining, Cfl-s1 for the floor; EN 1021 ignitability for upholstery	EN 13501-1; EN 1021
Fire safety provisions	Smoke-detector mount wired to the building alarm; provision for a sprinkler drop head or pass-through in the roof panel; door opens from inside without key or power at all times, electronic lock releases on power loss; main isolator reachable from outside	German building law (MBO / BauO Berlin), VdS CEA 4001:2024, host-building fire concept
Acoustics	Laboratory reports for the complete cabin and for the door; sound power of the air-conditioning unit at low and full speed	EN ISO 10140; ISO 23351-1

Area	Requirement	Standard / rule
Electrical / CE	CE under the Low Voltage Directive 2014/35/EU, EMC Directive 2014/30/EU, RoHS; Radio Equipment Directive for Wi-Fi, Bluetooth or NFC modules. CB-scheme reports (IEC 60335 / 62368-1 / 60598) from TUV, SGS or Intertek; EMC test of the complete cabin with all devices running; bill of electrical components with their own declarations; Schuko wiring. CE documentation of the air-conditioning unit including refrigerant type and EU F-gas requirements	EN 62368-1, EN 55032/55035, EN IEC 61000-3-2/-3-3, EN 60598; EU F-gas Regulation
Light sources	Photobiological safety report for every light source, including the red/NIR panel (with IR retinal-thermal class) and the virtual window; flicker data PstLM and SVM	EN 62471; Ecodesign 2019/2020
Air treatment	No ionisers, "plasma" or ozone devices. UV-C surface disinfection (254 nm, ozone-free) only while the cabin is locked and with a door interlock	UBA indoor-air commission; Directive 2006/25/EC; IEC 60335-2-65
Surfaces and textiles	PVC-free and phthalate-free surfaces; zipped waterproof washable mattress and pillow encasements; wipeable non-fabric surfaces for all shared contact items in the Health Pod; antimicrobial additives only with approved actives and labelling (or none)	EU Biocidal Products Regulation
Reset	Radar confirms the cabin is empty; full air change for the whole 20-minute reset; linen cassette exchangeable in under five minutes	HAVN operating concept
Sustainability	Recycled content of panels, foams and felts; materials separable and recyclable at end of life; single panels repairable or replaceable; energy consumption of the air-conditioning unit and lighting; pollutant testing of the finished cabin (formaldehyde, VOC, SVHC such as phthalates and flame retardants) as part of acceptance	REACH SVHC; AgBB; environmental certificates (e.g. ISO 14001, EPD) welcome

8. What matters most, and what can follow

Tier	Single and Double Cabin	Health Pod
Essential from the first cabin	Envelope and door to the acoustic targets (wall 52 dB, door 50 dB) with silencers on the air-conditioning penetrations; the Soundbox air-conditioning unit with CO2-controlled fresh air, 16 °C setting and the heat/condensate path; true blackout with hard-off relays; RGB+CCT lighting with the sunrise and bed-exit floor lights; two speakers and a microphone wired to a controller with an open API; radar and environment sensors with cabling; mattress encasement and linen cassette; prepared ceiling or wall position for the virtual window; the complete documentation of section 7. Double: air-conditioning sized for two, two reading lights	Single Cabin baseline plus: panel mounting frame at fixed distance and cabling; irradiance map and EN 62471 report; session timer and interlock; air-conditioning at full capacity during sessions; audio and light for the 20-minute protocols; wipeable surfaces; manufacturer's declared non-medical intended purpose
Can follow later	Virtual window (Circadian Sky 450 x 1200 portrait or OEM 46 mm edge-lit light box) scheduled as daylight dose and sunrise, off during sleep; opt-in lavender cartridge diffuser; water-loop bed-cooling pad option (dual-zone in the Double); weighted blanket as an option with its own wash cycle; opt-in sleep report with explicit consent; interlocked 254 nm UV-C reset pass; cove uplight and light matte ceiling	HRV/resonance-breathing audio with optional wearable integration; guided rest library; bright nap-exit light; vibroacoustic transducers in the bed framed as relaxation; opt-in FIR warm mat capped around 40-45 °C
Experimental	Rayleigh sun-and-sky skylight in a roof cut-out where host ceilings allow; far-UVC 222 nm only with ozone monitoring; in-house measurement of window versus no window	Massage bed; PEMF mat only with screening gate; cold exposure, oxygen and grounding excluded

Where a point is difficult in the Soundbox system, HAVN is glad to discuss alternatives; none of the points above is a knock-out criterion.

9. Points we would like to discuss

1. Overall: which of these points are standard for Soundbox today, which would need development, and where would you advise a different route? Every requirement can be adapted together.
2. Materials: which formaldehyde grade do your boards carry (E1 / E0 / ENF under GB/T 39600-2021), what are core, laminate, foam, felt and floor made of, and can you provide the REACH Appendix 14 chamber report and an AgBB VOC test on the assembled build-ups?
3. Fire: which fire reports do you hold today (GB 8624 or EN 13501-1 from an EU laboratory), and can you reach B-s1,d0 linings and Cfl-s1 floor?
4. Acoustics: what Rw do your wall and door achieve today (EN ISO 10140-2) and which ISO 23351-1 class? What would it take to reach 52 dB for the wall and 50 dB for the door with the air-conditioning penetrations silenced, and what has your best complete cabin measured so far?
5. Air-conditioning unit: cooling and heating capacity, fresh-air share and CO2 control, filter stages, temperature range (down to 16 °C), sound power of the noise-cancelling version at low and full speed, power consumption,

dimensions on the roof, refrigerant, and the heat and condensate path in a room-in-room installation.

6. Height: can the panel system deliver 2.40 m interior height while staying within 2.70 m total height including the unit?

7. Integration: can an 84 mm luminaire or a 46 mm light box sit within one 80 cm panel module without losing acoustic performance, and can the roof panel take a cut-out for a sun-and-sky unit? Do you have Guangdong partners for artificial-sky or virtual-window panels?

8. Lighting and controls: RGB+CCT with true-white channels, with measured spectra, melanopic values and flicker data per channel? Which controller do you propose, and does it have an open, documented API for the HAVN app?

9. Audio: two speakers with clean, good sound quality and one microphone per cabin, wired to the controller, with a hardware mute for the microphone?

10. Red-light panel: supplier, chip count, drive current, nine-point irradiance map at 20 cm, EN 62471 report including the IR retinal-thermal class, fan noise, surface temperature, timer and interlock design, declared intended purpose.

11. CE: will Soundbox supply CB-scheme reports, the whole-cabin EMC test and component declarations so that HAVN can hold the EU technical file as manufacturer of record?

12. Fire provisions: ceiling provision for sprinkler drop heads or pass-through, smoke-detector mount, fail-safe door release on power loss, external main isolator?

13. Reset: mattress encasement and linen-cassette fixing, interlocked UV-C module, radar occupancy sensor, controller platform and API?

14. Sustainability: recycled content, recyclability, repairability, energy consumption, and any environmental certificates you hold (for example ISO 14001 or environmental product declarations).

10. Selected references

Brown et al. 2022, recommendations for daytime, evening and night-time indoor light exposure (PLOS Biology). https://pubmed.ncbi.nlm.nih.gov/35298459/

WHO Europe, noise fact sheet (30 dB(A) in bedrooms at night). https://www.who.int/Europe/news-room/fact-sheets/item/noise

REHVA, European standardisation on indoor air quality (EN 16798-1 categories). https://www.rehva.eu/rehva-journal/chapter/european-standardisation-on-indoor-air-quality

DTU, how much ventilation is necessary to avoid disturbance to sleep. https://orbit.dtu.dk/en/publications/how-much-ventilation-is-necessary-to-avoid-disturbance-to-sleep/

Northwestern University on Zee et al. 2022 (PNAS), light exposure during sleep. https://news.northwestern.edu/stories/2022/03/close-the-blinds-during-sleep-to-protect-your-health

Windred et al. 2024, sleep regularity and mortality in 60,977 UK Biobank participants. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10782489/

Meta-analysis of daytime napping and cardiovascular disease, 21 cohorts. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11482734/

Canazei et al. 2017, artificial skylight in a windowless office (randomised, n = 100). https://www.sciencedirect.com/science/article/abs/pii/S036013231730344X

Journal of Building Engineering 2025, tunable artificial window, randomised crossover. https://www.sciencedirect.com/science/article/abs/pii/S2352710225015669

Gabel et al. 2013, dawn simulation after sleep restriction. https://orbi.uliege.be/bitstream/2268/171514/1/Gabel_CI_2013.pdf

Vanin et al., photobiomodulation and exercise performance and recovery, meta-analysis of 39 RCTs. https://pmc.ncbi.nlm.nih.gov/articles/PMC6546960/

Gimenez et al. 2023, double-blind RCT of 850 nm light exposure. https://pubmed.ncbi.nlm.nih.gov/36671752/

CIRS, REACH restriction on formaldehyde (entry 77). https://www.cirs-group.com/en/chemicals/the-use-of-formaldehyde-and-formaldehyde-releasing-substances-will-be-restricted-under-reach

Umweltbundesamt, AgBB evaluation scheme 2024. https://umweltbundesamt.de/sites/default/files/medien/4031/dokumente/agbb_evaluation_scheme_2024.pdf

NAMSA, MDR Annex XVI implementing regulation. https://namsa.com/resources/blog/mdr-implementing-regulation-annex-xvi-published/

Fagerhult, flicker and temporal light artefacts (PstLM, SVM). https://www.fagerhult.com/knowledge/technology/driver/flicker-and-tla/

Innerscene, Circadian Sky FAQ (84 mm plenum). https://www.innerscene.com/faq/circadian-sky

Seeed Studio MR60BHA2 60 GHz breathing and heartbeat radar module. https://openelab.io/cs/products/seeed-studio-xiao-60ghz-mmwave-human-breathing-heartbeat-sensor-mr60bha2

Nine Hours, sleep analysis report service. https://ninehours.co.jp/en/sleep-analysis-report